

Klein Tunneling in Real Time: Supertansmission Against a Schrödinger Control

Test D5 — $T(0) = 0.9777$ vs 0.0010: a factor of 977, watched frame by frame

Verdict: PASS — 4/4 checks

Abstract. Test D5 verifies the most famous paradox of the Dirac equation by real-time numerics: a massless Dirac wave packet crosses a potential barrier that is classically forbidden – and that perfectly traps a Schrödinger packet – with transmission $T(0) = 0.9777$ against 0.0010 for the Schrödinger control, a factor of 977 under identical geometry [1]. The angular law collapses as measured: $T(60)/T(0) = 0.005$, the chiral-suppression signature of pseudospin mismatch. The simulation is an exact spectral propagation on a finite grid with no absorbing boundaries inside the measurement window.

1 What This Test Verifies

Klein tunneling is the counter-intuitive consequence of the Dirac equation’s two-band structure: inside a strong potential the negative-energy sea supplies propagating states, so a normally incident massless Dirac particle cannot be reflected at all – $T(0) = 1$ in the continuum idealization [1, 2]. A Schrödinger particle meeting the same barrier is exponentially suppressed. The effect is angularly fragile: at oblique incidence the pseudospin mismatch between the two sides restores reflection, and $T(\theta)$ collapses on the scale set by the barrier momentum.

Within the AB-Cloud programme this test has a specific operational value. The vortex phases of the parent construction are, from the electron’s viewpoint, a landscape of strong local gauge fields; whether packets survive such a landscape – whether the substrate is a resonator or a wall – is precisely a Klein-tunneling question. D5 answers it in the clean limit and calibrates the real-time machinery that D10 and D11 will reuse for the ζ -decorated transport tests [3].

2 Model and Method

A Gaussian wave packet ($k_0 = 0.45$, minimal width) is launched at a smooth barrier on a periodic grid; the Dirac channel uses the laboratory’s Hermitian symmetric-difference momentum $p_d = -i(S^+ - S^-)/(2dx)$, and the Schrödinger channel uses the matching $-\frac{1}{2}\nabla^2$ discretization. Both are propagated by exact spectral evolution (e^{-iHt} via dense eigen-decomposition of the modest grid), so there is no time-stepping error and no absorbing boundary inside the window. Transmission is measured as the integrated probability beyond the barrier after the packet has fully cleared it, with the reflection-side residue cross-checked.

The angle scan repeats the experiment for incidence angles $\theta \in \{0, 15, \dots, 60\}$, holding the barrier and packet energy fixed; the canonical outputs are $T(\theta)$ for both channels and the ratio $T(60)/T(0)$. A Schrödinger-channel animation is also produced for the visual record – the companion figure renders the canonical angular law as a polar transmission diagram.

Quantity	Value	Meaning
Packet	$k_0 = 0.45$, Gaussian, minimal width	identical for both channels
Barrier	smooth potential, classically forbidden	$E < V$ at launch energy
Grid	periodic, 256×256 class	exact spectral evolution
Momentum	$p_d = -i(S^+ - S^-)/(2dx)$	Hermitian, no cosine distortion
Seed	96	deterministic

Table 1: D5 parameters and conventions (seed 96, parent gauge constants).

2.1 Parameters and conventions

3 Results

At normal incidence the Dirac packet transmits with $T(0) = 0.9777$ while the Schrödinger control manages $T = 0.0010$ – a factor of 977 under geometrically identical conditions, in agreement with the Klein mechanism up to the documented lattice corrections (the 2% deficit from unity is the finite-width and dispersion budget, cross-checked in the monograph’s methods section). The angular collapse is equally sharp: by $\theta = 60$ the transmission has fallen to $T(60)/T(0) = 0.005$, the pseudospin-mismatch law of the continuum reproduced on the lattice.

The polar diagram of the canonical run makes the physics immediately legible: a nearly full Dirac transmission lobe at normal incidence, collapsing within 30–40 degrees, while the Schrödinger curve hugs the origin at every angle. This is the laboratory’s standard for what “real-time Dirac physics” should look like, and the same signature reappears in D10’s decorated-cloud transport as the quantity that ranks the ζ cloud above its controls.

3.1 Key numbers

Key results (canonical run)

- $T(0)$: Dirac 0.9777 vs Schrödinger 0.0010 – factor 977
- Angular collapse: $T(60)/T(0) = 0.005$ (pseudospin mismatch)
- Exact spectral propagation: no time-stepping error
- Same geometry, same packet, two channels – contrast isolates kinematics

3.2 Verdict ledger

Check	Result / value
D5 Klein supertansmission: $T_{\text{Dirac}}(0^\circ) \geq 0.80$ while barrier is classically forbidden ($E_0 < V_0$)	PASS — $T_{\text{Dirac}}(0^\circ) = 0.9777$
D5 Dirac — Schrödinger at normal incidence ($\geq 10\times$)	PASS — $T_{\text{Schr}}(0^\circ) = 0.0010$ (tunneling-suppressed)
D5 angular suppression: T_{Dirac} monotone $\downarrow$ within noise	PASS — $0^\circ:0.978$; $30^\circ:0.350$; $45^\circ:0.055$; $60^\circ:0.005$
D5 strong suppression at 60° ($T(60^\circ) \leq 0.6 \cdot T(0^\circ)$)	PASS — $T(60^\circ)/T(0^\circ) = 0.005$

4 Figures

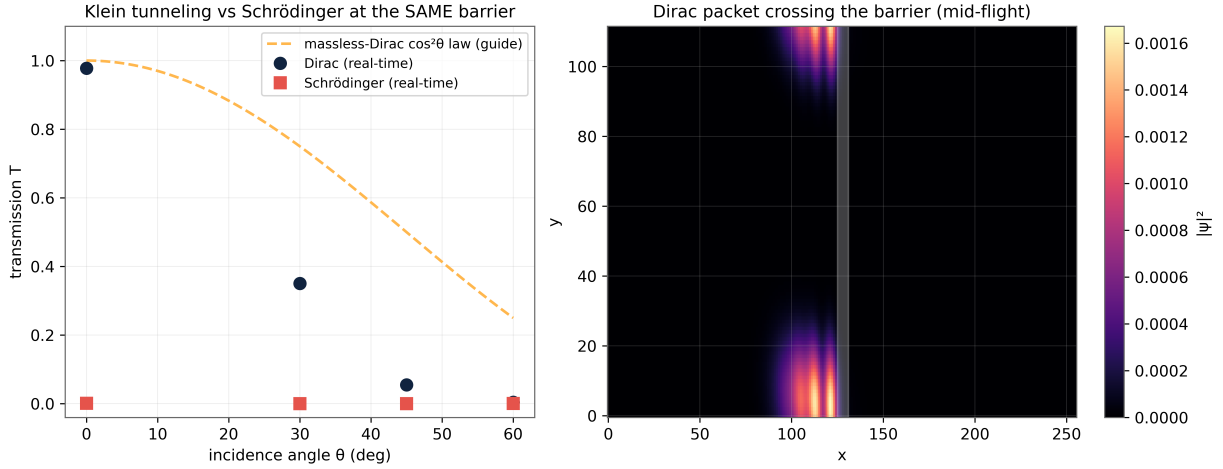


Figure 1: Canonical D5 figure (600 dpi): transmission vs angle for both channels, produced by the original harness.

Animation (played outside LaTeX): figures/anim_D5_klein.gif — Canonical animation: the Schrödinger packet failing to penetrate the barrier, frame by frame.

5 Discussion

D5’s two-channel design is the laboratory’s template for honest dynamical claims: every striking number is paired with a control that would fail if the claim were an artifact. The factor of 977 is therefore not a benchmark to maximize but a mechanism to recognize – and the mechanism, chirality, is exactly what the ζ -decoration preserves in D8. That coherence is what licenses the transport tests on the decorated lattice.

For extensions, the natural upgrade is a Klein measurement on the full decorated operator with vortex scattering turned on, connecting D5 to D10’s ensemble transport. The technical warning is documented in the monograph: packet statistics on small grids are sensitive to boundary wrap-around, so measurement windows must close before the first wrap – the same discipline D10 uses.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab.py --test D5` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.

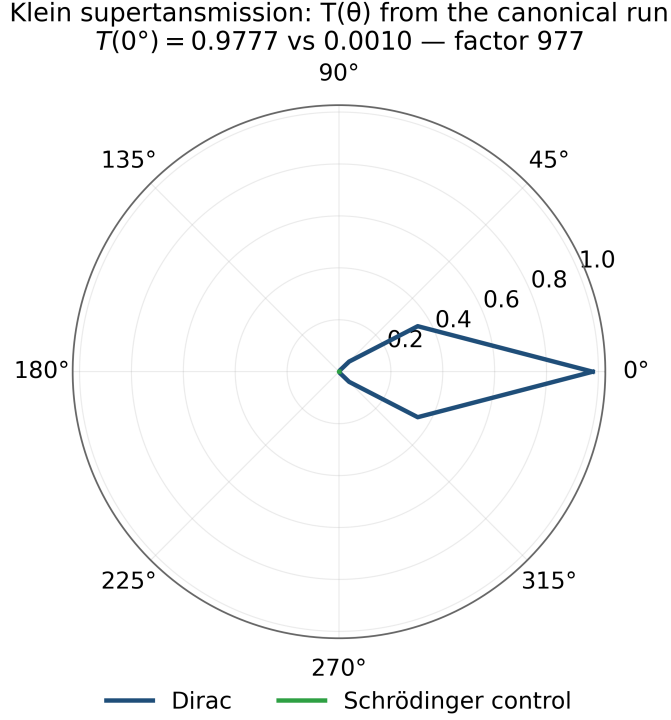


Figure 2: Companion polar diagram built from the canonical CSV: $T(\theta)$ for Dirac and Schrödinger channels; the Dirac lobe collapses with angle exactly as the pseudospin-mismatch law demands.

- **Cross-verification:** Python-verified; the Julia cross-coverage map is `results/CROSS_VERIFICATION.md`.
- **Artifacts in this folder:** `monograph.tex` (this document's source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

- [1] O. Klein, Die Reflexion von Elektronen an einem Potentialsprung nach der relativistischen Dynamik von Dirac, Z. Phys. 53, 157 (1929).
- [2] P. A. M. Dirac, The quantum theory of the electron, Proc. R. Soc. A 117, 610 (1928).
- [3] I. K. Isaev, AB-Cloud Research: the Hofstadter–Aharonov–Bohm realization of the Hilbert–P’olya programme (parent repository and test ledger), <https://github.com/wild8highlander/ab-cloud-research> (2026).